



G185HAN01.001 EC Report

AUO Display Plus

2023/09/11

G185HAN01.001 -> 005 EC Notice

Purpose :

Since Tcon IC (12304 Z03) will be EOL. In order to keep supply smoothly, ADP will launch a replaced model, including new cell, PCBA(Tcon IC 12303 Z09) for longevity support.

Replacement model :

G185HAN01.001 => G185HAN01.005 , new product specification need to be changed as below.

- EC Change
 - T con IC is changed from AUO 12304 Z03 to AUO 12303 Z09
 - Liquid Crystal is changed from ZAU-5396XX to LCT-16-1931
 - PCB layout

Verification schedule:

1. Verification samples will come out at the 2023/9.
2. RA test result : PASS
3. Customer Approval : 2023/12
4. Phase in => 2024/Q1

General Display Characteristics

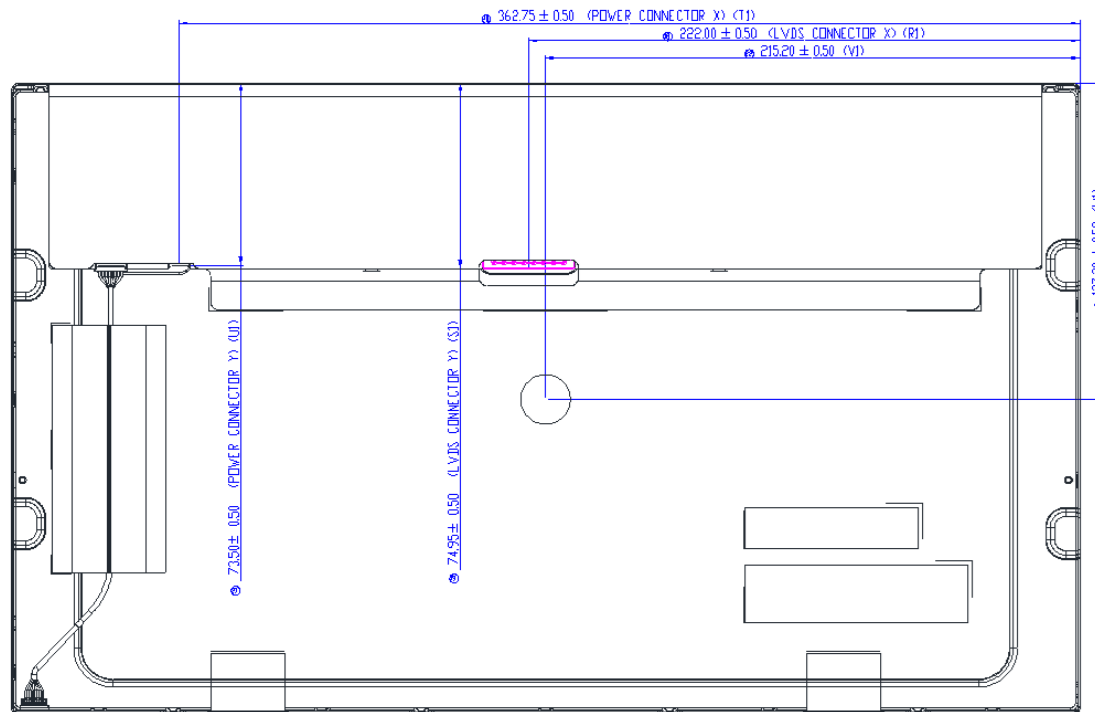
Panel supplier	MP (now)	EC model
Mode	AHVA	AHVA
Model name	G185HAN01.001	G185HAN01.005
Resolution	1920 x 1080	1920 x 1080
Active Area(mm)	408.96 x 230.04	408.96 x 230.04
Outline dimension(mm) (W x H x D)	430.4 x 254.6 x 12	430.4 x 254.6 x 12
Treatment	Antiglare (Haze 25%) / (3H)	Antiglare (Haze 25%) / (3H)
Contrast Ratio (Typ.)	1000:1	1000:1
Viewing Angle (CR≥10) (Typ.)	89/89/89/89	89/89/89/89
Response Time (Tr+Td) (Typ.)	20ms	20ms
Brightness (cd/m ²) (Typ.)	350	350
NTSC	72%	72%
Weight (g)	1200g	1200g
Interface	LVDS / True 8 bit	LVDS / True 8 bit
Operating Temperature	-20~70 °C	-20~70 °C
Storage Temperature	-20~70 °C	-20~70 °C
LED life time	50K	50K
LED driver board	Y	Y
Color Coordinates	x:0.299 ; Wy:0.315	x:0.299 ; Wy:0.315

EC Change

Items	Before GI85HAN01.001	After GI85HAN01.005
Tcon IC	AUO 12304 Z03	AUO 12303 Z09
Liquid Crystal	ZAU-5396XX	LCT-16-193I
PCB layout	Old	New

PCBA Change

- Keep outline and same mounting hole.



RA report-1

- PASS: Ref. G185HAN01.102

Product No	G185HAN01.1		Part No	97.18G03.102		Applicant/Dept		许奎/RRPBF3		Judgement		
Test Purpose	G185HAN01.102 DVT1-1 EA test					Application Date		2023/07/06		PASS		
						Input Date		2023/07/10				
RA No	D230707003		Product Stage	DVT		Issue Date		2023/07/11				
Test Item	Test Condition	Test Panel				Test Result						
THB	50°C 80% 500h	No	Chip_Id	Shipping_No		T0★	T100★	T200★	T300★	T500	Judgement	
		D2307070030001		113b4nk1klzzzs0102		OK	OK	OK	OK	OK	PASS	PASS
		D2307070030002		113b4nk1aezzzs0102		OK	OK	OK	OK	OK	PASS	
		D2307070030003		113b4nk1razzzs0102		OK	OK	OK	OK	OK	PASS	
HTO	70°C 500h	No	Chip_Id	Shipping_No		T0★	T100★	T200★	T300★	T500	Judgement	
		D2307070030004		113b4nk1rbzzzs0102		OK	OK	OK	OK	OK	PASS	PASS
		D2307070030005		113b4nk1jgzzzs0102		OK	OK	OK	OK	OK	PASS	
		D2307070030006		113b4nk1bmzzzs0102		OK	OK	OK	OK	OK	PASS	
LTO	-20°C 500h	No	Chip_Id	Shipping_No		T0★	T100★	T200★	T300★	T500	Judgement	
		D2307070030007		113b4nk1rmzzzs0102		OK	OK	OK	OK	OK	PASS	PASS
		D2307070030008		113b4nk1kazzzs0102		OK	OK	OK	OK	OK	PASS	
		D2307070030009		113b4nk1lazzzs0102		OK	OK	OK	OK	OK	PASS	
LTS	-20°C 500h	No	Chip_Id	Shipping_No		T0★	T100★	T200★	T300★	T500	Judgement	
		D2307070030010		113b4nk1bezzs0102		OK	OK	OK	OK	OK	PASS	PASS
		D2307070030011		113b4nk1tbzzzs0102		OK	OK	OK	OK	OK	PASS	
		D2307070030012		113b4nk1gazzzs0102		OK	OK	OK	OK	OK	PASS	
HTS	70°C 500h	No	Chip_Id	Shipping_No		T0★	T100★	T200★	T300★	T500	Judgement	
		D2307070030013		113b4nk1hhzzzs0102		OK	OK	OK	OK	OK	PASS	PASS
		D2307070030014		113b4nk1nbzzzs0102		OK	OK	OK	OK	OK	PASS	
		D2307070030015		113b4nk1nczzzs0102		OK	OK	OK	OK	OK	PASS	
TST	- 20°C/30min, 60°C/30min, 100 cycles judge 100cycles	No	Chip_Id	Shipping_No		C0★	C100★				Judgement	
		D2307070030016		113b4nk1bkzzzs0102		OK	OK				PASS	PASS
		D2307070030017		113b4nk1qbzzzs0102		OK	OK				PASS	
		D2307070030018		113b4nk1cmzzzs0102		OK	OK				PASS	

RA report-2

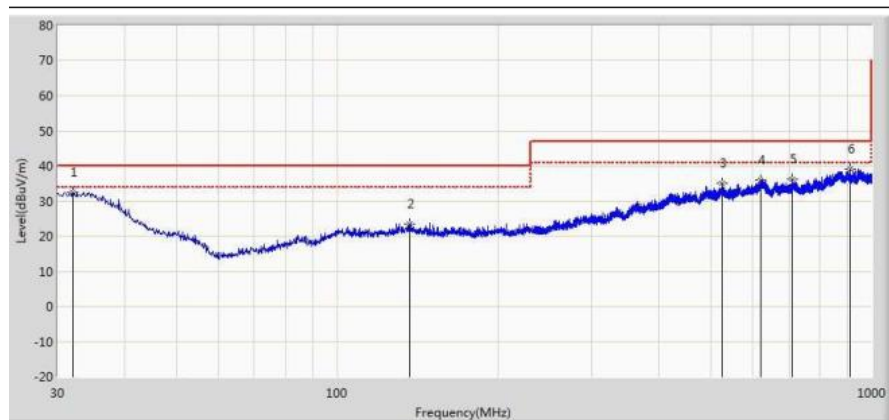
- PASS: Ref. G185HAN01.102

Optical Measurement Report													
Originator	chongfeng		Model/PCS			G185HAN01.1/5PCS			Date	2023/7/17			
Test Condition	Lum.,CR,Chromaticity,Uniformity,Gary Scale,V-T,View Angle,RT,CrossTalk,Flicker												
Test System	Detector	Inverter			DC Voltage					Warm up time			
FPM-520-2	SR-UL1R	NA			12V					30 min			
【 Test Reason】													
G185HAN01.102 DVT1-1 EA test													
【 Judgement】													
PASS													
Test Items	unit	PSS			23.5°C	23.5°C	23.5°C	23.5°C	23.5°C	-4σ	mean	+4σ	
					2023/7/16 8:48	2023/7/16 9:35	2023/7/16 10:02	2023/7/16 10:28	2023/7/16 11:26				
					3B4NK1PHZZZS010	3B4NK1RLZZZS010	3B4NK1TGZZZS010	3B4NK1BBZZZS010	3B4NK1UDZZZS010				
					G0300236O000W015	G0300236O000W015	G0300236O000W015	G0300236O000W015	G0300236O000W015				
		min	typ	max									
Center Brightness	cd/m ²	400	500		544.1	529.7	560.4	548.4	546.3	501.859	545.8	589.701	
Darkness (L0)					0.37	0.35	0.36	0.37	0.37	0.330	0.36	0.399	
Contrast Ratio(CR)	---	800	1000		1473.3	1510.0	1549.8	1468.7	1496.3	1368.712	1499.6	1630.509	
Brightness(5P Ave.)	cd/m ²				510.4	507.5	528.2	528.2	523.8	479.744	519.6	559.504	
Chromaticity													
	Wx	---	0.249	0.299	0.349	0.286	0.289	0.288	0.287	0.286	0.282	0.287	0.292
	Wy	---	0.265	0.315	0.365	0.301	0.304	0.301	0.300	0.300	0.295	0.301	0.307
	Rx	---	0.596	0.646	0.696	0.640	0.640	0.641	0.642	0.641	0.638	0.641	0.644
	Ry	---	0.283	0.333	0.383	0.332	0.332	0.331	0.331	0.332	0.331	0.332	0.332
	Gx	---	0.255	0.305	0.355	0.292	0.292	0.290	0.292	0.292	0.289	0.292	0.295
	Gy	---	0.565	0.615	0.665	0.599	0.601	0.601	0.599	0.599	0.595	0.600	0.605
	Bx	---	0.105	0.155	0.205	0.147	0.147	0.147	0.147	0.147	0.147	0.147	0.148
	By	---	0.010	0.060	0.110	0.071	0.071	0.070	0.069	0.070	0.067	0.070	0.074
	NTSC	---	---	72%	---	70.3	70.8	71.1	70.9	70.5	69.394	70.718	72.043
	CCT	---	---	---	---	8860	8501	8721	8873	8874	8120.619	8765.801	9410.984

EMI / ESD report

EMI

PASS



ESD

PASS

Contact Discharge			Air Discharge		
IEC Level	Test Voltage	Result	IEC Level	Test Voltage	Result
1	±2kV	A	1	±2kV	A
2	±4kV	A	2	±4kV	A
3	±6kV	B	3	±8kV	B
4	±8kV	B	4	±15kV	B



Tap Into The Possibilities